
GenDSR: Transferring Representations from Video Generation Models for Dynamic Spatial Reasoning

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Abstract

Dynamic spatial reasoning is essential for connecting visual intelligence to the physical world, yet remains challenging for vision-language models (VLMs), which must preserve scene state while tracking how spatial relations evolve over time. Meanwhile, video generation models (VGMs), trained to synthesize temporally coherent world dynamics, offer a rich source of spatiotemporal representations. We present *GenDSR*, a framework that transfers intermediate representations from VGM into VLM for dynamic spatial reasoning. Rather than treating a VGM feature as a monolithic prior, *GenDSR* constructs two complementary branches: a raw branch preserving holistic generative context and a delta branch emphasizing temporal variation. Task-conditioned tokenwise gates then fuse both signals and inject the result into the native visual representation. On DSR-Bench, *GenDSR* achieves **67.2%**, surpassing the previous best result and establishing a new state of the art.

1 Introduction

The physical environments in which intelligent systems operate are constantly changing: objects move, viewpoints shift, and spatial relationships evolve over time. Reasoning in such environments requires more than recognizing static scene states—it requires tracking and understanding their changes across time. Dynamic spatial reasoning, which relates observed motion to changes in position, orientation, distance, and inter-object relations, is therefore fundamental to intelligence in the physical world. Tasks such as navigation, manipulation, and autonomous decision-making depend not only on understanding the current spatial configuration, but also on anticipating how it changes (Zhou et al., 2026; Huang et al., 2026a).

Yet, despite substantial advances in vision-language models and video understanding, dynamic spatial reasoning remains far from reliable: recent benchmarks reveal persistent failures in integrating motion cues, disentangling camera and object motion, and maintaining temporally coherent spatial estimates (Liao et al., 2026; Zhou et al., 2025; Zhang et al., 2025b).

A prominent line of recent work enhances VLMs with spatial priors from pretrained 3D foundation models (Zhou et al., 2026; Zhang et al., 2026a), such as VGGT (Wang et al., 2025) and π^3 (Wang et al., 2026). While their latent features encode rich geometric structure, they are primarily learned for scene reconstruction rather than modeling temporal evolution, and rely on geometry-centric training data that are costly and difficult to collect at scale.

In contrast, video generation models (Wan et al., 2025; Yang et al., 2025c) provide a particularly suitable source of priors for dynamic spatial reasoning. Specifically, (1) **Spatial consistency**. To synthesize coherent videos under object and camera motion, these models must preserve scene structure and object identity across continuously changing viewpoints. Prior studies further suggest that their internal representations capture meaningful spatial cues, including camera pose and 3D structure, indicating spatial awareness beyond appearance-level semantics (Huang et al., 2026b;

Yang et al., 2026; Shen et al., 2026). (2) **Temporal dynamics.** Video generation explicitly requires modeling how visual content evolves over time, including object motion, camera movement, and interactions among entities. This objective naturally encourages representations sensitive to temporal changes that are essential for reasoning about evolving spatial relationships (Ho et al., 2022). (3) **Scalable learning.** Unlike geometry-centric models that depend on specialized multi-view data and geometric supervision, video generation models can learn directly from abundant real-world video collections. This provides a scalable source of both spatial and temporal knowledge without requiring costly 3D annotations or reconstruction labels. This naturally raises a question: can the representations learned by video generation models serve as dynamics-aware priors for spatial reasoning?

To this end, we propose *GenDSR*, a lightweight framework that transfers generative representations into VLMs for dynamic spatial reasoning. Our key insight is to expose two complementary views of the generative representation: its holistic spatiotemporal context and its explicit local variations. Accordingly, *GenDSR* constructs generative representations with two complementary branches. (1) **Raw branch.** We directly align the intermediate features of the video generation model with the VLM visual space, preserving rich spatial and spatiotemporal information about the scene. (2) **Delta branch.** We explicitly capture temporal variations by computing feature differences across neighboring observations, emphasizing changes induced by object motion, viewpoint shifts, and evolving spatial relationships. The two branches are jointly integrated through lightweight residual fusion, complementing the semantic representations of the original VLM with information about both spatial states and their temporal evolution.

On DSR-Bench (Zhou et al., 2026), *GenDSR* improves the Qwen3-VL-8B SFT baseline from 62.0% to 67.2%, while the 4B variant improves from 58.8% to 62.7%. Raw and delta representations independently improve the accuracy, and their combination yields the strongest overall performance. These results validate video generation models as a promising source of dynamic spatial priors for vision-language reasoning.

Our contributions are summarized as follows:

- We explore video generation models as a source of spatial and temporal priors for dynamic spatial reasoning, offering an alternative to geometry-centric representations.
- We propose *GenDSR*, which constructs two complementary representations from generative features: a raw representation of spatial states and a delta representation emphasizing temporal variations, and efficiently integrates both into VLMs.
- Experiments demonstrate consistent improvements in dynamic spatial reasoning and reveal the complementary roles of generative spatial context and temporal variations.

2 Related Work

2.1 Dynamic Spatial Reasoning

Spatial reasoning in vision-language models has progressed from understanding static layouts to reasoning about spatial configurations that evolve over time (Yang et al., 2025a; Zhou et al., 2026). Unlike static settings, dynamic spatial reasoning requires models to integrate evidence across frames, distinguish object motion from viewpoint changes, and maintain temporally coherent estimates of position, orientation, and inter-object relations. Recent studies show that even strong VLMs remain unreliable under these coupled spatial and temporal demands (Zhou et al., 2025; Huang et al., 2026a).

Existing methods improve dynamic spatial reasoning through task-specific post-training, explicit geometry, or structured internal representations. SpaceR (Ouyang et al., 2025) combines large-scale video spatial data with spatially guided reinforcement learning. DSR Suite (Zhou et al., 2026) formalizes dynamic spatial reasoning from monocular videos and introduces geometry-aware training data and a Geometry Selection Module that retrieves question-relevant information from pretrained 3D reconstruction priors. Building on geometry-centric representations, GeoSR (Zhang et al., 2026a) encourages VLMs to actively exploit geometry tokens through visual-token masking and adaptive geometric fusion. DynTrace (Gao et al., 2026) instead maintains geometry-informed object trajectories and structured dynamic traces at inference. Related work has also explored learning internal geometric or dynamic representations for reasoning. 3DThinker (Chen et al., 2025)

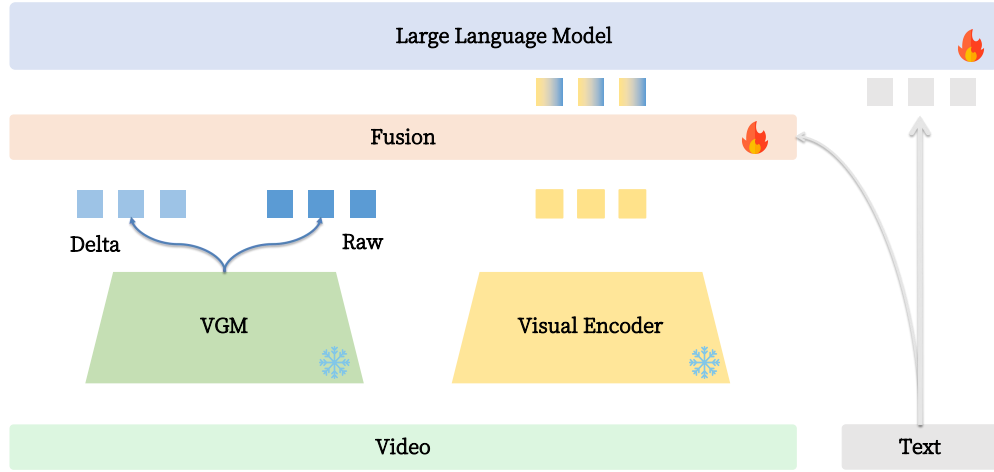


Figure 1: Overview of *GenDSR*. The frozen native visual encoder and video generation model process the same input video. We construct raw and delta representations from the VGM features and integrate them with the native visual representation through task-conditioned fusion before language-model reasoning. Snowflake and flame icons denote frozen and trainable modules, respectively.

aligns latent reasoning states with representations from a pretrained 3D foundation model, while 4DThinker (Chen et al., 2026) introduces dynamic latent imagery and reinforcement learning for spatiotemporal reasoning.

2.2 Video Generation Models for Spatial Reasoning

Recent evidence suggests that video generation models learn spatial structure beyond the appearance and motion statistics required for synthesis (Huang et al., 2026b; Shen et al., 2026). Moving from analysis to transfer, VEGA-3D (Wu et al., 2026) extracts intermediate video-diffusion features as implicit 3D priors and adaptively fuses them with MLLM visual tokens for scene understanding. DyVA (Zhang et al., 2025a) similarly repurposes a video diffusion model as a generative encoder, querying it with a single denoising step and injecting the resulting latents as auxiliary visual embeddings. GeoWorld-VLM (Gu et al., 2026) transfers geometry from a camera-conditioned world model through feature distillation. Another line of work turns generative models into explicit visual imagination. MindJourney (Yang et al., 2025b) couples a VLM with a controllable world model at test time and iteratively synthesizes views along model-proposed camera trajectories. Zhang et al. (2026c) systematically studies this generate-then-reason paradigm. World2VLM (Zhang et al., 2026b) instead uses trajectory-conditioned world-model generations as training-time supervision, distilling forward and inverse spatial reasoning into the VLM. Different from prior approaches that transfer generative representations as a unified auxiliary feature, *GenDSR* focuses on dynamic spatial reasoning and explicitly exposes two complementary views of the VGM representation: its raw spatiotemporal state and its local temporal variation. This decomposition allows the VLM to access temporal changes directly rather than relying on them to remain implicitly encoded in the raw generative feature.

3 Method

3.1 Overview

Given a video x , a question q , and an answer sequence y , a video-language model first encodes the video into visual representations and predicts the answer autoregressively. We denote the frozen

visual encoder by E_{vis} and the native visual merger by M_{vis} :

$$H = E_{\text{vis}}(x), \quad S = M_{\text{vis}}(H) \in \mathbb{R}^{N \times d}, \quad (1)$$

where N denotes the number of visual tokens and d is the hidden dimension of the language model. We associate these tokens with a spatiotemporal grid $G = (T_v, H_v, W_v)$, such that $N = T_v H_v W_v$. Our goal is to enrich the native visual representation S with complementary knowledge from a frozen video generation model (VGM), while preserving the original visual pathway of the VLM.

Figure 1 provides an overview of *GenDSR*. We extract intermediate representations from the pretrained VGM and construct two complementary representations from them: a **raw branch** that preserves holistic generative context and a **delta branch** that explicitly emphasizes temporal variations. Both representations are aligned to the native VLM token grid and projected into the language-model hidden space. A lightweight, task-conditioned fusion module then determines their contributions at each visual-token position and injects the resulting representation as a residual immediately before the language model.

The following sections introduce the raw-delta representation transfer in Sec. 3.2, the adaptive fusion mechanism in Sec. 3.3, and the training objective in Sec. 3.4.

3.2 Raw-Delta VGM Representation Transfer

Generative feature extraction. Unlike a conventional visual encoder, a diffusion-based VGM operates through a latent denoising process. We first encode the observed video with the frozen VGM VAE:

$$z_0 = E_{\text{vae}}(x). \quad (2)$$

For timestep τ , we draw $\epsilon \sim \mathcal{N}(0, I)$ and apply the native VGM scheduler’s noising operator $\mathcal{N}_\tau$:

$$z_\tau = \mathcal{N}_\tau(z_0, \epsilon), \quad (3)$$

where $\mathcal{N}_\tau$ follows the VGM’s native noise schedule. We then perform one forward pass through the frozen denoising transformer under the empty-prompt condition $c_\emptyset$. Let $\mathcal{D}_\ell(z_\tau; \tau, c_\emptyset)$ denote the intermediate activation at block ℓ , after patch embedding and propagation through the preceding transformer blocks. We extract

$$Z = \mathcal{D}_\ell(z_\tau; \tau, c_\emptyset) \in \mathbb{R}^{T_g \times H_g \times W_g \times d_g}, \quad (4)$$

where the first three dimensions form the native VGM feature grid and d_g is the VGM channel dimension. The empty-prompt condition prevents an external text prompt from contributing semantic information. This extraction requires only a single denoising-network forward at a fixed timestep, rather than an iterative denoising trajectory.

The VGM grid (T_g, H_g, W_g) generally differs from the VLM grid G . We use one shared, non-parametric alignment operator $\mathcal{A}_G$ for both branches:

$$\mathcal{A}_G : \mathbb{R}^{T_g \times H_g \times W_g \times d_g} \rightarrow \mathbb{R}^{N \times d_g}. \quad (5)$$

It interpolates the temporal axis to T_v , applies adaptive average pooling to the target spatial grid (H_v, W_v) , and flattens the aligned grid into N tokens. The two branches use projectors with the same architecture but independent parameters. For $k \in \{r, d\}$, each projector is implemented as

$$P_k(U) = W_{k,2} \text{ GELU}(W_{k,1} \text{ LN}_k(U) + b_{k,1}) + b_{k,2}, \quad (6)$$

where the first linear layer maps d_g to d dimensions and the second operates in the d -dimensional language-model space.

Raw branch. The raw VGM representation retains the rich spatiotemporal context learned through video generation. We first align its native spatiotemporal grid with that of the VLM and then project the aligned features into the language-model hidden space:

$$R = P_r(\mathcal{A}_G(Z)) \in \mathbb{R}^{N \times d}, \quad (7)$$

where P_r is a learnable feature projector. Rather than replacing the native representation S , the resulting feature R provides complementary spatial and spatiotemporal context acquired from generative pretraining.

Delta branch. While the raw representation contains rich contextual information, temporal changes may remain entangled with appearance and scene content. We therefore explicitly expose temporal variations in the VGM feature space by computing temporal differences between neighboring features:

$$(\Delta Z)_1 = \mathbf{0}, \quad (\Delta Z)_t = Z_t - Z_{t-1}, \quad t = 2, \dots, T_g, \quad (8)$$

where T_g denotes the temporal resolution of the VGM representation. Prepending the zero feature preserves the original temporal length, so ΔZ contains the same T_g positions as Z . Importantly, temporal differencing is performed in the native VGM feature space before spatiotemporal alignment, so temporal differences are defined directly on the native VGM representation rather than on resampled features.

The resulting temporal differences are subsequently aligned and projected:

$$D = P_d(\mathcal{A}_G(\Delta Z)) \in \mathbb{R}^{N \times d}, \quad (9)$$

where P_d is a learned projector. The delta representation D emphasizes temporal variations associated with object motion, viewpoint changes, and evolving spatial relationships, complementing the holistic context retained by R .

3.3 Task-Conditioned Tokenwise Fusion

The usefulness of raw and delta information can vary across questions and visual locations. For example, questions about static spatial configurations may rely primarily on scene context, whereas questions about motion or changing relationships may benefit more from temporal variations. A fixed global mixture therefore cannot adapt to the diverse requirements of dynamic spatial reasoning.

Let $\mathcal{I}_p$ index the valid prompt-side text tokens, excluding visual tokens, target-response tokens, and padding, and let $e_j \in \mathbb{R}^d$ be the corresponding input embedding from the VLM language embedding layer. For our multiple-choice prompts, $\mathcal{I}_p$ includes both the question and its answer-option text. We obtain the task representation by mean pooling:

$$Q = \frac{1}{|\mathcal{I}_p|} \sum_{j \in \mathcal{I}_p} e_j \in \mathbb{R}^d, \quad \tilde{Q} = \mathbf{1}_N Q^\top \in \mathbb{R}^{N \times d}. \quad (10)$$

The input embeddings are detached only when constructing Q , which prevents the gate from introducing an auxiliary gradient path through the prompt representation. We broadcast Q over the visual-token grid and condition the gate on the native visual tokens, both VGM branches, and the resulting task context. Specifically, we form

$$U = [\text{LN}_s(S), \text{LN}_r(R), \text{LN}_d(D), \text{LN}_q(\tilde{Q})] \in \mathbb{R}^{N \times 4d}, \quad (11)$$

$$w_k = h_k(U) = UW_k + b_k, \quad k \in \{r, d\}, \quad (12)$$

where $[\cdot]$ denotes feature concatenation and h_r, h_d are independent linear heads from $\mathbb{R}^{4d}$ to $\mathbb{R}$. The unconstrained coefficients allow each branch to be independently amplified, suppressed, or sign-adjusted. We initialize W_r and W_d to zero and their biases to $b_r = 1.0$ and $b_d = 0.1$, respectively.

The fused generative residual is then computed as

$$F = w_r \odot R + w_d \odot D, \quad (13)$$

$$S' = S + F, \quad (14)$$

where $\odot$ denotes tokenwise scaling with broadcasting over the hidden dimension. Only visual-token representations are modified; text-token embeddings remain unchanged. The residual is injected after the native visual merger and immediately before the language model, allowing the generative features to complement rather than replace the native VLM representation.

3.4 Training Objective

The augmented visual tokens S' together with the question q condition the standard autoregressive supervised fine-tuning objective:

$$\mathcal{L}_{\text{SFT}} = - \sum_{i=1}^L \log p_{\Theta}(y_i \mid y_{<i}, S', q), \quad (15)$$

Table 1: **Dynamic spatial reasoning on DSR-Bench.** Accuracy (%) is reported for 13 subtasks and overall accuracy.

Models	A.Dis	A.Dir	A.Ori	A.Spd	A.SpdC	A.DirP	R.Dis	R.Dir	R.Ori	R.Spd	R.SpdC	R.DirP	N-Temp	Avg. ↑
<i>Proprietary Models</i>														
GPT-5	21.1	41.5	48.7	34.5	33.3	34.7	17.2	44.3	41.9	21.2	25.0	30.9	26.7	30.8
Gemini-2.5-Pro	20.0	44.6	53.6	27.3	38.7	30.5	23.2	32.9	43.2	17.1	28.5	27.9	34.3	31.7
<i>Spatial Reasoning Models</i>														
VLM-3R	28.2	27.6	31.7	42.8	38.7	33.0	34.4	23.8	30.8	22.2	26.4	29.1	35.0	31.4
VG-LLM	55.2	32.3	58.5	57.1	51.6	32.2	56.0	36.3	32.0	30.3	32.1	29.1	27.9	38.4
DSR Suite-Model	87.0	73.8	84.1	73.8	72.0	35.5	75.8	76.1	77.7	60.6	37.1	35.1	46.4	58.9
GeoSR	88.0	84.4	87.8	83.3	78.5	41.0	79.5	83.0	83.8	76.5	48.9	39.8	52.9	66.1
4DThinker	84.3	79.7	82.3	84.0	67.0	47.8	67.0	82.8	79.5	62.9	44.2	41.0	48.5	62.0
<i>Ours (Based on Qwen3-VL-4B)</i>														
Qwen3-VL-4B	34.1	31.7	29.5	37.5	14.3	30.4	22.5	29.5	35.4	20.8	19.6	20.6	35.7	27.7
+SFT	89.0	71.4	82.1	72.5	62.6	43.5	77.5	76.1	79.7	49.0	38.4	38.8	46.0	58.8
+SFT+Raw	89.0	73.0	85.9	68.8	65.9	46.1	77.5	78.4	82.3	59.4	41.3	33.3	48.1	60.2
+SFT+Raw+Delta	91.5	76.2	91.0	66.2	74.7	42.6	77.5	81.8	81.0	75.0	50.7	37.6	43.0	62.7
<i>Ours (Based on Qwen3-VL-8B)</i>														
Qwen3-VL-8B	23.5	24.6	42.6	29.7	27.9	33.8	18.1	28.4	34.5	24.2	22.1	27.9	33.5	28.7
+SFT	89.0	82.5	88.5	78.8	70.3	47.8	74.8	83.0	86.1	63.5	42.0	36.4	43.4	62.0
+SFT+Raw	89.0	87.3	89.7	77.5	80.2	43.5	76.6	88.6	86.1	83.3	58.7	38.2	41.3	65.8
+SFT+Raw+Delta	89.0	82.5	92.3	77.5	80.2	44.3	76.6	90.9	83.5	84.3	53.6	38.2	52.0	67.2

where L denotes the length of the target answer and Θ contains all trainable parameters. Both the VGM, including its VAE and denoising transformer, and the native VLM visual encoder remain frozen. We optimize the language model, the native visual merger, the two branch projectors, and the fusion module.

4 Experiments

4.1 Experimental setup

Benchmark and metrics. We evaluate on DSR-Bench (Zhou et al., 2026), a multiple-choice benchmark for fine-grained dynamic spatial reasoning from monocular videos. It contains 13 subtasks: six spatial attributes—distance, direction, orientation, speed, speed comparison, and direction prediction—are evaluated under absolute and relative viewpoints, together with a non-template subset. We report accuracy for each subtask and overall sample-level accuracy.

Implementation details. Our default model uses Qwen3-VL-8B-Instruct (Bai et al., 2025) as the VLM and frozen Wan2.1-T2V-1.3B (Wan et al., 2025) as the VGM. We extract VGM activations offline from zero-indexed block 20 under an empty-text condition and diffusion timestep of 300. We train on the DSR-train split for one epoch with a global batch size of 32 and a learning rate of 2×10^{-7} . The VGM and native VLM visual encoder remain frozen. We use AdamW with a cosine learning-rate schedule, a warmup ratio of 0.03, and weight decay of 0.01. All experiments are conducted on $8 \times A100$ GPUs.

4.2 Results and Analysis

Table 1 evaluates whether VGM representations improve over strong public and proprietary systems. *GenDSR* with Qwen3-VL-8B reaches **67.2%**, surpassing the previous best GeoSR (Zhang et al., 2026a), which uses the 3D foundation model π^3 (Wang et al., 2026), establishing a new state of the art on DSR-Bench. The gain is not explained by supervised fine-tuning alone: Qwen3-VL-8B SFT obtains 62.0%, raw transfer raises accuracy to 65.8%, and the full raw-delta method reaches 67.2%. The 4B backbone shows the same ordering, with raw-delta fusion reaching 62.7% compared with 58.8% for SFT. These cross-scale results show that the transferred representation improves both backbones under the protocol.

4.3 Ablation studies

Raw and delta branches. Table 2 evaluates the two transferred signals. Raw and delta features independently improve over SFT, while their combination performs best, supporting their complementarity.

Table 2: **Raw–delta components.**

Variant	Raw	Delta	Acc
Base	–	–	28.70
SFT	–	–	62.00
Raw only	✓	–	65.80
Delta only	–	✓	64.32
<i>GenDSR</i>	✓	✓	67.20

Table 3: **VGM block.**

Block	Acc
5	65.46
10	65.39
15	66.64
20	67.20
29	66.43

Table 4: **VGM timestep.**

Timestep	Acc
No noise	65.60
$t = 100$	65.04
$t = 300$	67.20
$t = 600$	66.92
$t = 900$	66.43

VGM feature block. Table 3 varies the extraction layer while keeping the raw–delta pathway, diffusion timestep, and training protocol fixed. Block 20 performs best, while both shallower and deeper features are weaker. This indicates that transferable evidence is concentrated in an intermediate-to-late representation rather than increasing monotonically with network depth.

Diffusion timestep. Table 4 examines how the diffusion timestep shapes the transferred representation. Performance is lower for the clean and lightly noised states, peaks at an intermediate timestep, and then gradually declines as the corruption becomes stronger. This non-monotonic trend suggests that an intermediate noising state may better balance the preservation of input evidence with the VGM’s learned denoising prior.

5 Conclusion

We introduce *GenDSR*, which extracts intermediate representations from a frozen video generation model and transfers them into a VLM through complementary raw and delta branches. Experiments on DSR-Bench show that generative representations consistently improve VLM performance. These results suggest that video generation models can serve not only as synthesis models, but also as a valuable source of representations for reasoning about the evolving physical world.

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